The Price of Fairness, Complexity, and Incentives in No-Numeraire Combinatorial Trade-Intent Auctions

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Abstract. Blockchain trade-intent auctions currently intermediate billions of dollars in trades per month. They are combinatorial because a solver executing several intents jointly can net opposing flows and save on fees. Yet they have no numeraire. There is no common asset in which to redistribute the resulting surplus. The deployed fix builds an endogenous fairness benchmark from the bids and executes a batched bid only if it delivers to every covered trader at least that trader's standalone competitive outcome. We give a worst-case theory of this design using a common scale parameter, the complementarity factor g, together with liquidity $\theta = 1 - \lambda$. The model and fairness benchmark are due to Canidio and Henneke [8]. Our contribution is the worst-case approximation, competitive, and incentive layer on top.
The price of fairness is exactly $\min(g, 1/\theta)$, tight for all parameters and independent of the number of orders and of prices, with a dichotomy between g (no numeraire) and 1 (full numeraire). Batch size is not the right parameter. The right refinement is a per-leg coverage floor α, under which the exact law is $\min(g, \alpha + (1-\alpha)/\theta)$. Within the order-price abstraction of UDP, the rate-realizable floor-vector frontier for equilibrium support of efficient allocations is exactly $g = 1$, by native no-numeraire arguments. At $g = 1$ the price system is precisely the fairness benchmark. The price of strategyproofness is $\Theta(1 + \log g)$ for a single multi-minded solver and for many single-minded solvers, via two posted-price mechanisms and a convex profit-curve technique. The remaining multi-agent multi-minded cell is a partial frontier. It admits $O(N \log g)$ and has a serial-demand $\Omega(\min(N, g))$ barrier. Online, the price splits. Against realized lifetime floors even randomization pays exactly g, while with static public floors the residual problem is one-way search over the spread, deterministic $\Theta(\sqrt{g})$ and randomized $\Theta(\log g)$. Across these results, the same scale-free parameter g recurs in both the informational and the incentive price of the spread.

Keywords: Combinatorial auctions · Price of fairness · Mechanism design · Blockchain · Competitive analysis.

1 Introduction

Blockchain "trade-intent" auctions let users state a desired exchange of one asset for another and delegate execution to competing agents called *solvers*. Such systems collect intents off-chain and auctions them in batches, in the spirit of frequent-batch market design [6]. These markets are economically significant. Recent figures put monthly volume on the order of billions of dollars.[1] They are also *combinatorial*. A solver executing several intents together can net opposing flows and save on fees, producing more value than executing them separately. This batching logic is central in MEV-aware market design too [12,7]. What makes them unlike the combinatorial auctions of the literature is the absence of a *numeraire*. Users may demand any asset for any other, there is no common unit of account, and assets are illiquid, so the extra value created by batching cannot simply be paid out and shared.

A deployed intent-auction protocol uses the *fair combinatorial auction*. It constructs a fairness benchmark *from the bids themselves*. Each trader's standalone competitive outcome defines a floor, and a batched bid is executed only if it delivers at least that floor to *every* covered trader. This sidesteps the impossibility of sharing batching gains by never allowing a batch to make any trader worse off than individual competition would. The equilibrium behavior of this design was analyzed by Canidio and Henneke [8], who identify a tension between the fairness guarantee and the value returned to traders. Thus the model and benchmark are theirs. Our contribution is the worst-case approximation layer around them. We ask how much welfare the fairness filter costs, when competitive equilibria exist, what strategyproofness costs, and how these prices behave online and under manipulation.

A common scale. We organize such a theory around a recurring structural quantity. For a solver with batch value $V_i(S)$ on a set S of intents and standalone values $V_i(\{j\})$, the *complementarity factor* is the least $g \geq 1$ with $V_i(S) \leq g \sum_{j \in S} V_i(\{j\})$ for all i, S. It measures how much more a batch can be worth than the sum of its parts. The benchmark floor of order j is $c_j = \max_i V_i(\{j\})$, and liquidity is $\theta = 1 - \lambda \in [0, 1]$, where λ is the haircut on moving value across assets ($\theta = 0$ is the pure no-numeraire regime, $\theta = 1$ restores a numeraire). We show that g, together with θ, is the natural scale in a wide range of worst-case quantities, which gives the design a unified analysis rather than a collection of unrelated bounds.

The role of g is deliberately worst-case. The theorems do not require a deployment to estimate the structural complementarity of every batch. The empirical section only shows that the corresponding observable spreads and supports are real in deployment.

Contributions. We ask what price several design requirements carry, each in terms of g (and liquidity θ). Each item below leads with tight characterizations (exact or asymptotically tight). Partial frontiers and peripheral extensions are marked as such.

[1] Order-of-magnitude figure from the anonymized deployment diagnostics in Appendix F. Precise deployment identifiers are omitted for anonymity.

1. **Price of fairness (Section 3).** The welfare ratio between the unconstrained optimum and best fair outcome is exactly $\min(g, 1/\theta)$, tight and order/price-independent, with endpoints $\{g, 1\}$. Under coverage floor α the exact law is $\min(g, \alpha+(1-\alpha)/\theta)$. The binding instance is an asymmetry at the repair/drop indifference point, not maximal complementarity.
2. **Equilibrium existence and the core (Section 4).** A Walrasian/UDP equilibrium supporting an efficient allocation exists for every rate-realizable floor-vector instance iff $g = 1$, with obligations equal to the fairness floors. The deployed filter is the singleton-core constraint, the individual allocation lies in the tight g-core, and the no-numeraire core can be nonempty when the transferable core is empty.
3. **Price of strategyproofness (Section 5).** Endogenous best-bid fairness conflicts with DSIC already on one order, and online realized floors force ratio g even randomized. In the public historical-floor regime, strategyproofness costs $\Theta(1 + \log g)$ for many single-minded solvers and for one multi-minded solver. As further separations, independent geometric option pricing admits an $\Omega(g/\log^2 g)$ loss, and the multi-agent multi-minded cell remains a partial frontier. It has an $O(N \log g)$ mechanism and a serial $\Omega(\min(N, g))$ barrier.
4. **Online and strategic prices (Section 6).** Online fair settlement pays exactly g against realized floors. With static public floors the residual one-way search problem is deterministic $\Theta(\sqrt{g})$ and randomized $\Theta(\log g)$. The individual first-price restriction is smooth but leaves a latent-batch obstruction for the full game. Appendix E records a complete but peripheral benchmark-manipulation extension with feedback $g \mapsto g/(1 - \mu)$.

Fair winner determination is NP-hard but FPT in directed pairs. Section 7 reports diagnostics. The worst-case theory is independent of them. Table 1 collects the results.

Related work. The model and the equilibrium analysis of the fair combinatorial auction are due to Canidio and Henneke [8]. We complement their 2×2 equilibrium study with a worst-case, approximation-theoretic analysis. Recent intent-market and solver-competition models give nearby strategic context [10], and batch trading has been proposed as a response to AMM arbitrage, LVR, and sandwich attacks [7]. Combinatorial auctions are classical [11], but almost always with a numeraire and quasilinear utilities. Our setting has neither, putting it closer to mechanism design without money [17] and no-transfer impossibility phenomena in MEV-aware fee mechanisms [1]. The price of fairness was introduced for divisible resources [4] and developed for indivisible goods under proportionality, envy-freeness, and MMS [9,3,2]. Here it scales with production complementarity and liquidity, not agent count. Equilibrium existence draws on assignment markets [19] and package assignment [5]. Our smoothness bound uses Syrgkanis–Tardos [20] and intrinsic robustness [18]. Online search instantiates one-way trading [13]. Order-flow-auction models under PBS provide related strategic context [16].

Table 1. Worst-case prices as functions of the complementarity factor g and liquidity θ (N is the number of strategic solvers and α is coverage). "exact" marks literal equalities or fully tight characterizations. "asymptotically tight" marks $\Theta(\cdot)$ bounds.

Quantity	Result	Status
Price of fairness	$\min(g, 1/\theta)$, endpoints $\{g, 1\}$	exact
coverage refinement	$\min(g, \alpha + (1-\alpha)/\theta)$, size-irrelevant	exact
Walrasian/UDP existence	rate-realizable class with frontier $g = 1$	exact
Core	filter = singleton-core, g-core nonempty and tight	exact
Winner determination	NP-hard, FPT by directed-pair count	complexity classification
Strategyproofness (screening)	rand. $\Theta(1 + \log g)$, det. exact $\sqrt{g}$	tight/asympt. tight
single-/multi-minded	both $\Theta(1 + \log g)$, indep. geometric loss $\Omega(g/\log^2 g)$	asympt. tight / separation
multi-agent multi-minded	$O(N \log g)$ / serial $\Omega(\min(N, g))$	partial (serial barrier)
Online (realized floors)	exactly g, even randomized	exact
Online (static floors)	det. $\Theta(\sqrt{g})$, rand. $\Theta(\log g)$	asympt. tight
Price of anarchy	individual-only $\leq e/(e-1)$, full conj. $\Theta(g)$	partial (latent batches)
Contamination	threshold $\kappa_1 \geq v\beta$ deters, $g \mapsto g/(1-\mu)$	App. E

2 Model

There are n trade intents (*orders*) and a set of *solvers*. A solver i submitting a batched bid on a set S specifies, for each $j \in S$, a *raw output* $y_{i,S,j} \geq 0$ in j's requested asset (trader-normalized units), before any within-batch rebalancing. The batch value is $V_i(S) = \sum_{j \in S} y_{i,S,j}$, realized only if the batch executes. Delivery is batch-specific. The same solver may have different raw outputs for order j in different batches, so $V_i(S)$ is not additive in general. The standalone value is $V_i(\{j\}) = y_{i,\{j\},j}$, and the floor is $c_j = \max_i V_i(\{j\})$. Reports are *binding*. A solver cannot claim raw output or delivered value it cannot produce, so relevant misreports are downward. Welfare $W = \sum_j w_j$ is an additive social objective over trader-normalized delivered scores, not a transferable monetary surplus.

Definition 1 (Complementarity factor). *The instance has complementarity factor $g \geq 1$ if $V_i(S) \leq g \sum_{j \in S} V_i(\{j\})$ for every solver i and every S. Thus g caps how much batching amplifies standalone value. The case $g = 1$ is additive.*

Definition 2 (Fairness floor and fair allocation). *The floor of order j is $c_j := \max_i V_i(\{j\})$, the value of its best standalone bid, and $C := \sum_j c_j$. An allocation is fair if every allocated order receives $w_j \geq c_j$. Batched bids violating this for some covered order are filtered.*

An allocation selects winning bids covering disjoint order sets. Its welfare is $W = \sum_j w_j$. Let OPT be the maximum welfare ignoring fairness and FAIR the maximum over fair allocations. The *price of fairness* is PoF $:= \sup \mathrm{OPT}/\mathrm{FAIR}$ over a stated class.

Definition 3 (Liquidity and feasible delivery). *Within a single winning bid a solver may convert value across its traders at haircut $\lambda \in [0,1]$. Write $\theta := 1-\lambda$ for liquidity. Moving one unit out of one trader's asset yields θ units in another. For delivered vector w, let $D^+ = \sum_{w_j > y_{i,S,j}}(w_j - y_{i,S,j})$ and $D^- = \sum_{w_j < y_{i,S,j}}(y_{i,S,j} - w_j)$. Delivery is feasible iff $D^+ \le \theta D^-$. Thus a starved leg may receive $w_j > y_{i,S,j}$, funded by over-served legs. $\theta = 1$ is value-preserving and $\theta = 0$ forbids cross-trader repair. Conversion is per-bid.*

This per-bid model excludes cross-settlement inventory. Richer liquidity only eases repair, so $\min(g, 1/\theta)$ is a worst-case ceiling.

The deployed mechanism additionally requires uniform clearing prices per directed token pair (*uniform directed prices*, UDP), forcing all orders on one directed pair into a single winning bid. UDP is vacuous for the constructions in Section 3, which use distinct pairs. Sections 5–6 study public-benchmark screening and online abstractions without imposing UDP. Section 4 uses UDP as its obligation abstraction, and UDP later reappears as the source of fixed-parameter tractability. This is the order-level analogue of the uniform clearing price in frequent batch auctions [6], specialized to the trade-intent model of Canidio and Henneke [8]. Full proofs of all numbered results are in the appendix.

Regime map.

Result	Floors	Repair	UDP	Solvers
Thms. 1–2	standalone/given	yes, θ	no	any
Thm. 3	endogenous	no	yes	many
Thm. 4	current-report	no	no	many
Thms. 5–6	public/historical	no	no	screening / 1 multi / many single
Thm. 7	public/historical	no	no	N multi
Prop. 3–Thm. 8	realized/static	no	no	online
Thm. 9	given	no	no	many

Except in §3, repair is not part of feasible deviations.

3 The Price of Fairness

The first price is informational. Once the benchmark floors are fixed by standalone competition, how much welfare can the componentwise fairness filter destroy? The answer is governed exactly by g and liquidity θ.

Theorem 1 (Price of fairness, exact). *For every $\theta \in [0,1]$ and $g \ge 1$, over all instances with liquidity θ and complementarity factor at most g,*

$$\mathrm{PoF}(\theta, g) = \min\bigl(g,\ 1/\theta\bigr),$$

tight, and independent of the number of orders and of prices ($1/\theta := \infty$ at $\theta = 0$).

Proof (Proof idea). Upper bound. Process each bid of an unconstrained optimum independently. For a locally undominated OPT bid, replacing it by singleton floor allocations would not improve welfare, so $\gamma = V/C_S \geq 1$. Definition 1 gives $\gamma \leq g$. Its deficit is $D = \sum_{j \in S}(c_j - y_{i,S,j})^+ \leq C_S$. Either drop the batch and settle floors, delivering $C_S = V/\gamma$, or repair starved legs by a feasible delivery that saturates the conversion budget of Definition 3. Clearing D loses $D(1-\theta)/\theta$, so whenever repair is feasible,

$$\frac{V - D(1-\theta)/\theta}{V} \geq 1 - \frac{1-\theta}{\theta\gamma} \geq \theta.$$

Dropping covers low gains by V/g. Repair covers high gains by $V/(1/\theta)$. Summing over disjoint optimal bids gives the bound. *Lower bound.* A two-order instance with one starved leg, calibrated to $\gamma^* = \min(g, 1/\theta)$, forces FAIR $= c$ against OPT $= \gamma^* c$. The binding instance is an extreme asymmetry, not maximal complementarity. □

Corollary 1 (Dichotomy). *At the endpoints, the pure no-numeraire auction pays the full factor g, while a full numeraire removes the fairness cost entirely.*

A natural attempt to add a third parameter fails, and identifies the right one.

Proposition 1 (Batch size is not the parameter). *With* $\mathrm{PoF}(\theta, g, d)$ *the price restricted to bids covering at most d orders,*

$$\mathrm{PoF}(\theta, g, 1) = 1, \qquad \mathrm{PoF}(\theta, g, d) = \min(g, 1/\theta) \quad (d \geq 2).$$

Say an instance has *coverage* $\alpha \in [0, 1]$ if every bid satisfies $y_{i,S,j} \geq \alpha c_j$ on every covered order. No leg is worse than α times the benchmark.

Theorem 2 (The coverage law). *Write $\mathit{\Gamma}(\theta, \alpha) := \alpha + (1-\alpha)/\theta$, with $\mathit{\Gamma}(0, \alpha) = \infty$ for $\alpha < 1$ and $\mathit{\Gamma}(0, 1) = 1$. Over instances with liquidity θ, complementarity at most g, and coverage α,*

$$\mathrm{PoF}(\theta, g, \alpha) = \min\big(g, \mathit{\Gamma}(\theta, \alpha)\big),$$

tight on two-order instances.

The quantity $\mathit{\Gamma} - 1 = (1-\alpha) \cdot \frac{1-\theta}{\theta}$ is the largest relative per-leg shortfall times the conversion loss per unit of delivered deficit. The price of fairness prices the worst starved leg, discounted by liquidity. The endpoint $\alpha = 1$ is exactly the componentwise filter-passing under which fairness is free. This refinement also explains why batch size disappears from the theorem. What matters is not how many orders are bundled, but how much one covered leg can fall below its own standalone benchmark before the batch's surplus can be converted back to repair it.

4 Equilibrium Existence and the Core

The second question is structural. When does the same filter coincide with a competitive equilibrium or a cooperative stability notion rather than merely a feasibility rule? Here $g = 1$ is the threshold. This is not a restatement of the gross-substitutes existence theory. Gross substitutes support Walrasian equilibria with prices and transfers over items [15,14], whereas our frontier is about componentwise obligations in traders' requested assets. Throughout this section, equilibrium feasibility is the native UDP obligation model. An executable deviation must meet each obligation componentwise, equivalently $\theta = 0$ for deviations. The liquidity repair operator of Definition 3 is used for the price-of-fairness comparison, not for defining executable UDP deviations.

We work in the order-price abstraction of UDP. A directed-rate system induces order-level obligations $w_j \geq 0$ that any solver serving j must deliver, and the single property used is that w_j depends only on the order. A *UDP equilibrium supporting the efficient allocation* is a rate system and an allocation in which every order is served at exactly its w_j, every solver's executed portfolio maximizes its total profit $\sum_{S\in\mathcal{P}}\sum_{j\in S}(y_{i,S,j} - w_j)$ over feasible portfolios $\mathcal{P}$ (those with $y_{i,S,j} \geq w_j$ for every executed bid S and covered order j), and produced value equals OPT. This definition deliberately separates prices from payments. The quantity w_j is an obligation in order j's normalized delivered units, not a transfer that can compensate another trader. That separation is what makes the frontier different from the transferable package-assignment theorem.

Lemma 1 (Batch domination under $g = 1$). *If $g = 1$ then at any rates every executable batched bid is weakly dominated by the sub-portfolio of its positive-profit singletons: $\sum_{j\in S}(y_{i,S,j} - w_j) = V_i(S) - \sum_j w_j \leq \sum_j (V_i(\{j\}) - w_j)^+$. Hence every solver has an individual-only best response. The one use of UDP is that w_j is the same in a batch and in a singleton.*

The lemma is the reason $g = 1$ is special. If batching is never more productive than the sum of singletons, then any profitable executable batch contains enough profitable singleton mass to replace it. The no-numeraire feasibility constraint only strengthens this conclusion: some batch deviations that would be available with transfers are simply not executable componentwise. Conversely, once $g > 1$, overlap matters. A solver can create value on a pair without creating enough value on either order alone to support order-level obligations. The three-cycle lower bound exploits exactly this gap. Each pair wants one obligation low for feasibility and another high for incentive compatibility, and the cycle leaves no price vector that satisfies all three demands.

Theorem 3 (Existence frontier is $g = 1$). *Within the order-price abstraction of UDP, restricted to instances whose floor vector c is rate-realizable, a UDP equilibrium supporting an efficient allocation exists for every instance with complementarity at most g iff $g = 1$. For $g = 1$, the obligations $w_j = c_j$ form a best-bid-fair equilibrium. For every $g > 1$, some rate-realizable instance admits no such equilibrium. Solvers have no capacity constraints.*

Proof (Proof idea). Case $g = 1$. Set $w_j = c_j$ and assign each order to a best producer. Winners earn zero margin, losers nonpositive, and Lemma 1 kills batch deviations. Produced value is $C = \mathrm{OPT}$, and the price vector coincides with the fairness benchmark. *Case* $g > 1$. The obstruction is native. Non-existence with transfers does not imply it here, since feasibility only *removes* deviations. The hard instance is a three-cycle of overlapping pair-batches with vector $(1+\frac{\delta}{2}, 1+\frac{\delta}{2})$ over floors 1. The three orders lie on distinct directed pairs, so the unit floor vector is rate-realizable. Supporting one efficient pair forces an obligation on the third order low enough to keep that pair feasible. The adjacent unchosen pair is then both executable and must be unprofitable. Splitting on whether the shared obligation is above or below its singleton floor yields either $\delta \leq 0$ or a componentwise-feasibility contradiction. Full details are in Appendix B. □

The filter also has a cooperative reading. A bid *blocks* an allocation at factor κ if some feasible delivery gives every covered order strictly more than κw_j. The κ-core forbids this.

Proposition 2 (Core placement). *(i) Best-bid fairness is exactly immunity to singleton blocking at factor* 1*. The deployed filter is the singleton-coalition core constraint. (ii) The individual allocation* $w_j = c_j$ *lies in the* g*-core for every* θ*, and* g *is the least universal blocking-immunity factor. (iii) There are instances whose transferable-utility core is empty while the* $\theta = 0$ *core is nonempty. The separation is driven by overlap, as in the integrality gap.*

Thus the fairness filter is not an ad hoc admissibility test. It is the smallest coalition-stability condition visible when coalitions are restricted to orders' own requested assets. The multiplicative g-core statement says that even when exact support fails, the individual allocation is never blocked by more than the same complementarity factor that drives Section 3. This is one of three appearances of a single phenomenon. The no-numeraire constraint *removes deviations* (VCG payments become infeasible, batch deviations are dominated at $g = 1$, and componentwise blocking is weak). Whether the $\theta = 0$ core is always nonempty is open. Scarf's route needs balancedness, broken by the same overlap geometry.

Complexity. Fair winner determination is NP-hard (from 3-dimensional matching) but fixed-parameter tractable in the number P of directed asset pairs. Under UDP, all orders on the same directed pair settle atomically. A feasible solution either assigns the entire directed-pair block to one bid or leaves it unserved. Thus every fair bid is a weighted subset of the P atoms. Dynamic programming over pair masks (or equivalently weighted set packing on P items) tries 2^P states and scans the bid list in each state, giving $2^{O(P)} \cdot \mathrm{poly}(|\mathcal{B}|)$. A deployment cap on pairs per batch thus makes the problem tractable in the intended regime. This is another way in which g and the token-pair graph play complementary roles. The parameter g controls the loss from fairness and incentives, while P controls the search space left by the uniform-price constraint.

5 The Price of Strategyproofness

The third price is strategic. Truthfulness is not studied under the original endogenous benchmark, because the preceding structure creates a trilemma. Current-report floors, ex-post fairness, and VCG-style strategyproofness cannot coexist without transfers. This places the problem in the no-money mechanism-design tradition [17], with a blockchain-specific impossibility flavor close to post-MEV fee mechanism design [1].

Throughout this section, incentive compatibility means *one-sided dominant-strategy implementability under binding reports.* Upward misreports are infeasible by settlement, and "universal truthfulness" refers to posted-demand mechanisms whose menu is independent of the solver's behavior.

There are three benchmark regimes. First, *current-report* floors are the original endogenous ideal, but they are self-referential. VCG-style compensation is infeasible because the no-numeraire constraint gives no asset in which to make the Clarke payment. More sharply, best-bid fairness and dominant-strategy truthfulness already conflict on one order:

Theorem 4 (Best-bid fairness is incompatible with strategyproofness). *Call a mechanism non-wasteful if it allocates an order whenever some reported standalone value is positive, and value-respecting if it allocates to a solver with maximal reported value. No feasible, non-wasteful, value-respecting, DSIC mechanism guarantees fairness at the endogenous best-bid benchmark* $c_j = \max_i r_{ij}$*. This holds already for one order. Under DSIC, the enforceable benchmark collapses to the runner-up level.*

Second, *realized lifetime* floors are operationally stronger, but as Section 6 proves, online fairness then admits no ratio better than g even with randomization. A solver settling a batch can be undercut by a later standalone bid that raises a covered order's own floor. Third, *public delayed or historical* floors sever this self-reference while preserving the spread g as the operative parameter. This is the deployed-safe regime, not a hypothetical substitute. A deployment-safe benchmark is already a public historical window. We therefore analyze this exogenous-benchmark regime. The price of strategyproofness PoSP is the welfare ratio between FAIR and the best universally truthful fair mechanism, and solvers are single- or multi-minded with private values passing the filter. Under endogenous floors the relevant question is not the price of strategyproofness but its non-existence.

The canonical screening family of the regime is already nontrivial for one strategic solver:

Theorem 5 (Canonical screening family). *Over the screening family* R_g *(one strategic solver, public floors), the price of strategyproofness is* $\Theta(1 + \log g)$*. Every randomized DSIC mechanism pays at least* $\frac{g \ln g}{2(g-1)} \geq \frac{\ln g}{2}$*, a geometric posted-price mechanism achieves* $e(\lceil \ln g \rceil + 1)$*, and the best deterministic mechanism is exactly* $\sqrt{g}$*.*

The lower bound uses a model-native technique. A one-sided envelope inequality (downward incentive compatibility alone forces monotone utilities, so the utility derivative dominates the execution probability a.e.) is integrated against a V^{-2} kernel. In the screening family a type V can only underreport. If $p(V)$ is execution probability and $u(V)$ interim utility, downward IC gives

$$u(V) \geq u(V') + (V - V')p(V') \qquad (V' \leq V),$$

so $u'(V) \geq p(V)$ a.e. The scale-free integral forces logarithmic loss. The deterministic optimum balances the two errors at threshold $\sqrt{g}$. Random geometric posted levels give the matching upper bound up to constants. Two mechanisms lift the guarantee to the full game.

Mechanism model. In the exogenous-benchmark regime the floors c_j are public. The public geometry consists of feasible option sets S and their floor masses $C_S = \sum_{j \in S} c_j$. A solver's private type specifies, for each option S, an executable filter-passing production vector $y_{S,j} \geq c_j$. We summarize it by the scalar value $V_S = \sum_{j \in S} y_{S,j} \in [C_S, gC_S]$. Any total delivery $D \in [C_S, V_S]$ can clear all floors componentwise by scaling surplus. Deliver $w_j = c_j + \alpha(y_{S,j} - c_j)$ for $\alpha = (D - C_S)/(V_S - C_S)$ (with the obvious convention when $V_S = C_S$). The mechanisms below post only as a function of public C_S and a public random seed, and each solver responds by demanding a disjoint collection of options rather than reporting values. Settlement verifies the promised componentwise delivery. Universal truthfulness is in this posted-demand sense. For every seed, demanding a utility-maximizing executable collection is dominant because the menu is independent of that solver's behavior. The posted quantity $C_S t$ is not a monetary payment. It is a required delivered-score threshold for option S, in the covered orders' requested assets. The profit $V_S - C_S t$ is retained raw value, not cash.

Theorem 6 (Two routes to $\Theta(1 + \log g)$). *(i) For arbitrarily many single-minded solvers with public floors, an eligibility-posted packing mechanism is universally truthful and $\Theta(1 + \log g)$-competitive. (ii) For one multi-minded solver of arbitrary structure, a coupled-menu mechanism posts all options at prices $C_S\, t$ for a single random multiplier $t = e^k$ and lets the solver demand any profit-maximizing disjoint collection. It is universally truthful and $\Theta(1 + \log g)$-competitive. The natural independent geometric per-option analogue can instead pay $\Omega(g/\log^2 g)$.*

The multi-minded bound rests on a *profit-curve* technique. The solver's profit $p(t)$ over disjoint collections is convex and piecewise linear, delivered mass is $t\,(-p'(t))$, and a geometric grid telescopes $p(1)$ against expected welfare. The single random multiplier is essential. The independent geometric analogue lets a multi-minded solver take only the cheapest member of a family of substitutes, losing the collection value that fairness must compare against. The remaining cell is both multi-agent and multi-minded:

Theorem 7 (Multi-agent multi-minded). *Random-priority coupled menus are universally truthful with* $\mathbb{E}[W] \geq \mathrm{FAIR}/\big(N(e-1)(\lceil \ln g \rceil + 1) + 1\big)$, *i.e.* $\Theta_N(1 + \log g)$ *for a constant number* N *of strategic solvers. Conversely every serial-demand mechanism, meaning a report-independent ex-ante service order, arbitrary per-identity multiplier distributions, and each agent demanding a profit-maximizing collection, pays* $\Omega(\min(N, g))$, *even with geometry-aware ordering. So any* N*-independent* polylog(g) *mechanism must break the serial-demand format.*

The barrier is proved by a *symmetrized swarm.* All identities share the same public geometry (all singletons plus the grand set), differing only in private values, so no report-independent order can tell the batcher from the blockers. The telescoping identity

$$\sum_i (1 - q_i) \prod_{s \prec i} q_s = 1 - \prod_s q_s \leq 1$$

then places the batcher behind an expected blocker.

These results should be read as a trichotomy. Single-minded competition is handled by posted eligibility and packing. One multi-minded solver is handled by a coupled menu and profit-curve telescoping. Many multi-minded solvers expose a genuine serial-demand barrier. The open question is whether a mechanism that is not serial-demand can remove the dependence on N while keeping the logarithmic dependence on g.

6 Online and Robustness

The remaining results explain why deployed systems replace current-report floors by historical, public, and rate-limited ones.

Online floors. With realized lifetime floors, online fairness can be attacked by a later standalone bid that raises a covered order's own floor. This is the online analogue of the endogenous-benchmark impossibility in Section 5. The solver competes against a future revision of the fairness constraint itself.

Proposition 3 (Realized floors force a factor g). *The deterministic online competitive ratio of fair settlement is exactly* g.

Theorem 8 (Two online prices). *(i) Against realized lifetime floors with almost-sure fairness, randomization does not help. The competitive ratio is exactly* g. *(ii) With static public floors on single-contention instances, the residual problem is one-way search over the spread, with deterministic ratio* $\Theta(\sqrt{g})$ *and randomized ratio* $\Theta(\log g)$. *Ratio-threshold greedy mechanisms still pay* $\Omega(g)$.

Thus Theorem 8 is the operational reason for the exogenous-benchmark regime in Section 5. Once floors are fixed, the residual problem is one-way search over width g [13].

Strategic play. For the first-price game, measured in produced welfare PW rather than delivered welfare, the individual-only restriction is smooth in the standard composable-mechanisms sense [20,18]:

Theorem 9 (Individual competition is smooth). *Every coarse correlated equilibrium of the per-order-decomposable individual-only game has* $\mathbb{E}[\mathrm{PW}] \geq (1-\frac{1}{e})\mathrm{OPT}_{\mathrm{ind}}$*, hence price of anarchy at most* $e/(e-1)$ *relative to* $\mathrm{OPT}_{\mathrm{ind}}$ *and at most* $\frac{e}{e-1}\, g$ *relative to the unrestricted optimum. The full-game price of anarchy is conjectured* $\Theta(g)$*. The obstruction is latent batches that execute only under a deviation.*

Benchmark robustness. Benchmark manipulation is itself a priced game. A sufficient injection-cost floor deters it, and steady contamination feeds back as $g \mapsto g/(1-\mu)$, closing the loop with every floor-calibrated bound (Appendix E).

7 Deployment Diagnostics (Motivation)

An anonymized trace from a deployed intent-auction system contains 802,816 post-adoption solver competitions over eight months. On the sanity-filtered USD subsample it reports \$54.69B covered notional and a \$5.13M visible accounting gap (0.94 bps). The visible-spread tail is nontrivial. Across all auctions, $\widehat{G}^{\mathrm{obs}} > 2$ in 25.3% and $\widehat{G}^{\mathrm{obs}} > 4$ in 20.2%. For small-notional auctions, $\widehat{G}^{\mathrm{obs}} > 2$ in 63.1%. Appendix F defines the score variables, filters, and replay diagnostics. When the denominator is positive, the parser computes $\widehat{G}^{\mathrm{obs}}_a = Score_{1,a}/Score^{\mathrm{other}}_{2,a}$, where $Score^{\mathrm{other}}_{2,a}$ is the best positive valid score after removing the winning solver, not a singleton floor mass. Thus the spread is not an estimate of g, but it gives a conditional lower envelope. Let S_a be the winning batch and $C_a = \sum_{j \in S_a} c_j$ its floor mass. If $Score^{\mathrm{other}}_{2,a} = \eta_a C_a$, then $\eta_a \widehat{G}^{\mathrm{obs}}_a = Score_{1,a}/C_a \leq g_a$, because $Score_{1,a} \leq V_{\mathrm{win}}(S_a) \leq g_a \sum_{j \in S_a} V_{\mathrm{win}}(\{j\}) \leq g_a C_a$. The trace does not observe C_a, so η_a is an auxiliary-audit ratio. Without certifying $\eta_a \geq 1$ we do not read $\widehat{G}^{\mathrm{obs}}$ as an unconditional lower bound on g. These diagnostics motivate the parameter regime. They neither estimate structural g nor test point predictions. The bounds are tight as worst-case functions of g and hold for whatever g a deployment exhibits. The trace also records P and a reserve diagnostic. In that diagnostic, 97.1% of auction-weighted cell-windows are vulnerable under a naive local hard reserve, motivating inherited, quantile-based, delayed, and rate-limited benchmarks.

8 Conclusion

A recurring scale, the complementarity factor g with liquidity θ, organizes the worst-case fairness, equilibrium, complexity, incentive, and online prices of the no-numeraire fair combinatorial auction. Two phenomena recur. No-numeraire feasibility removes some deviations while disabling transfers, and one geometric family prices both informational and incentive costs of the spread.

Three questions remain open, each with an identified obstruction: full-game price of anarchy (conjectured $\Theta(g)$, with the latent-batch obstruction), polylogarithmic mechanisms for multi-agent multi-minded solvers without N dependence (the serial-demand barrier), and nonemptiness of the $\theta = 0$ core (Scarf balancedness, broken by overlap).

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A Full Proofs for Section 3

Proof (Proof of Theorem 1). Endpoint. If $\theta = 0$, repair is unavailable; dropping every OPT batch to the singleton floor allocation loses at most the factor g, and the lower-bound construction below attains g. Hence assume $\theta > 0$ for the repair calculation.

Upper bound. Fix an unconstrained optimum and process its winning bids independently; bids cover disjoint order sets, so the repaired welfares add. Take a winning bid with floor mass $C_S = \sum_{j\in S} c_j > 0$ (a bid with $C_S = 0$ has all floors zero and is already fair), value $V = \sum_{j\in S} y_j$, where $y_j := y_{i,S,j}$, gain $\gamma = V/C_S$, and deficit $D = \sum_{j\in S}(c_j - y_j)^+$. By local optimality of OPT, we may assume $V \geq C_S$ (otherwise replacing this bid by the singleton floor allocation on S improves welfare), so $\gamma \geq 1$; Definition 1 gives $\gamma \leq g$. Two fair uses:

Repair. Move value from over-served legs to starved ones. The total surplus available is $s = V - (C_S - D) = V - C_S + D$, and clearing the deficit costs D/θ units of surplus (haircut $\lambda = 1 - \theta$). This is feasible iff $V - C_S \geq D\frac{1-\theta}{\theta}$, and then the delivery saturating the conversion budget of Definition 3 has welfare $f = V - D\frac{1-\theta}{\theta}$. Since every raw output value is nonnegative, $(c_j - y_j)^+ \leq c_j$ and therefore $D \leq C_S$. Together with $V - C_S = (\gamma-1)C_S$, feasibility holds whenever $\gamma - 1 \geq \frac{1-\theta}{\theta}$, i.e. $\gamma \geq 1/\theta$; then

$$\frac{f}{V} = 1 - \frac{1-\theta}{\theta\gamma}\frac{D}{C_S} \geq 1 - \frac{1-\theta}{\theta\gamma} \geq \theta = \frac{1}{1/\theta}.$$

Thus repair gives $f \geq V/(1/\theta)$ whenever $\gamma \geq 1/\theta$, and hence $f \geq V/\min(g, 1/\theta)$ in the repair regime.

Drop. Settle every covered order at its floor: $f = C_S = V/\gamma$. For $\gamma \leq \min(g, 1/\theta)$ this is $\geq V/\min(g, 1/\theta)$.

Either way each bid retains a $1/\min(g, 1/\theta)$ fraction of its value; summing, $\text{FAIR} \geq \text{OPT}/\min(g, 1/\theta)$.

Lower bound. Let $\gamma^* = \min(g, 1/\theta)$. Two orders on distinct pairs, $c_1 = 0$, $c_2 = c$; one batched bid with $y_2 = 0$, $y_1 = \gamma^* c$, so $V = \gamma^* c$, $C_S = c$, and the batching solver's standalone values are $0, c$, so Definition 1 holds as $\gamma^* \leq g$. The unconstrained optimum runs this batch, delivering $\gamma^* c$ (all on order 1), so $\text{OPT} = \gamma^* c$. Fair repair of order 2 to its floor c requires withdrawing c/θ from y_1; if $\gamma^* = 1/\theta$ this exhausts y_1 exactly (delivered welfare c), and if $\gamma^* = g < 1/\theta$

the batch is irreparable (welfare c from the individual allocation). In both cases FAIR $= c$ and OPT $= \gamma^* c$, so OPT/FAIR $= \gamma^*$. Independence of the number of orders and of prices is immediate from the construction. $\square$

Proof (Proof of Proposition 1). The two-order family of Theorem 1 already has $d = 2$ and attains $\min(g, 1/\theta)$, and the upper bound never uses d; for $d = 1$ only singletons exist, so OPT = FAIR = C. $\square$

Proof (Proof of Theorem 2). Endpoint. If $\theta = 0$ and $\alpha < 1$, repair is unavailable and the drop argument gives ratio g; if $\alpha = 1$, every admitted bid already clears all floors and the ratio is 1. Hence assume $\theta > 0$ below.

Upper bound. Write $\Gamma = \Gamma(\theta, \alpha) := \alpha + (1-\alpha)/\theta$, with the endpoint convention from the theorem. As above, coverage caps the deficit by $D \le (1-\alpha)C_S$. Repair is feasible when $\gamma - 1 \ge (1-\alpha)\frac{1-\theta}{\theta} = \Gamma - 1$, and then $f = V - \frac{D(1-\theta)}{\theta} \ge V\bigl(1 - \frac{(1-\alpha)(1-\theta)}{\theta\gamma}\bigr) \ge V/\Gamma$ for $\gamma \ge \Gamma$ (monotone in γ, equality at $\gamma = \Gamma$); dropping gives $V/\gamma \ge V/\Gamma$ for $\gamma \le \Gamma$ and V/g for $\gamma \le g$. Hence FAIR $\ge$ OPT$/\min(g, \Gamma)$. *Lower bound.* Two orders, $c_1 = 0$, $c_2 = c$; batched bid with $y_2 = \alpha c$ (coverage saturated) and $y_1 = (\gamma^* - \alpha)c$, $\gamma^* = \min(g, \Gamma)$; the batching solver's standalone values are $0, c$, so Definition 1 holds as $\gamma^* \le g$. Repairing order 2 needs $(1-\alpha)c/\theta$ from y_1; if $\gamma^* = \Gamma$ then $y_1 = (\Gamma - \alpha)c = \frac{(1-\alpha)c}{\theta}$ exactly, and if $\gamma^* = g < \Gamma$ the batch is irreparable. Either way FAIR $= c$, OPT $= \gamma^* c$. $\square$

B Full Proofs for Section 4

Proof (Proof of Lemma 1).

$$
\begin{aligned}
\sum_{j \in S}(y_{i,S,j} - w_j) &= V_i(S) - \sum_{j \in S} w_j \\
&\le \sum_{j \in S} V_i(\{j\}) - \sum_{j \in S} w_j \\
&\le \sum_{j \in S} (V_i(\{j\}) - w_j)^+.
\end{aligned}
$$

The middle step uses $g = 1$. Each positive-profit singleton has $V_i(\{j\}) > w_j$, hence is executable; a non-executable batch is not a deviation. The obligation w_j is the same whether j is served in the batch or individually, which is the only property of UDP used. $\square$

Proof (Proof of Theorem 3). (a), $g = 1$. Set $w_j = c_j$ and assign each order j to a solver attaining $V_i(\{j\}) = c_j$. The winner earns zero margin per order; any losing singleton has $V_i(\{j\}) - c_j \le 0$; and Lemma 1 kills batch deviations (no positive-profit singletons). With $g = 1$, any allocation produces at most $\sum_{(i,S)} V_i(S) \le \sum_j c_j = C$, attained here, so produced value is $C =$ OPT, all delivered. The point $w_j = c_j$ is the best-bid floor, so the equilibrium is fair (the trader-optimal core point of the underlying assignment market [19], to which Lemma 1 reduces the economy).

(b), $g > 1$. We argue natively; non-existence with transfers does not transfer, as feasibility only removes deviations. Fix $\delta \in (0, 2(g-1)]$; orders $1, 2, 3$ on distinct pairs, so the floor vector $c_j = 1$ is rate-realizable. A rival has standalone value 1 on each ($c_j = 1$), and three solvers each holding one batched bid on a pair $\{j, j'\}$ with vector $(1+\frac{\delta}{2}, 1+\frac{\delta}{2})$ (value $2+\delta \le 2g$) and standalone values 1. The efficient allocation runs one batch, say $\{1, 2\}$, plus order 3 individually (produced $3+\delta$). Suppose rates w support it. Order 3 is served by a standalone solver of value 1, so $w_3 \le 1$; orders $1, 2$ are served by the batch, so $w_1, w_2 \le 1 + \frac{\delta}{2}$. The batch $\{2, 3\}$ is then executable, and equilibrium requires it unprofitable against its solver's individual alternatives: $(2+\delta) - w_2 - w_3 \le (1-w_2)^+ + (1-w_3)^+$. If $w_2 \le 1$ the right side is $(1-w_2) + (1-w_3)$, giving $\delta \le 0$; if $w_2 > 1$ it is $1 - w_3$, giving $w_2 \ge 1 + \delta > 1 + \frac{\delta}{2}$, contradicting feasibility. No supporting rates exist.□

Proof (Proof of Proposition 2). (i) If $w_j < c_j$, the standalone bid attaining c_j blocks j at factor 1; if $w_j \ge c_j$ for all j, no singleton strictly improves. (ii) Blocking $w_j = c_j$ at factor g needs delivery $> g\, c_j$ to every $j \in S$, totalling $> g\, C_S$, while any feasible delivery from a bid totals $\le V_i(S) \le g\, C_S$ (haircuts only lose value; if all floors in S are zero then $V_i(S) = 0$ and strict blocking is impossible). Tightness: a batch with vector $(g'c, g'c)$ over floors (c, c), $g' \in (\kappa, g]$, blocks at any $\kappa < g$. (iii) Three-cycle floors $(1, 1, 1)$, pair vectors $(1+\frac{\delta}{2}, 1+\frac{\delta}{2})$: summing the three transferable core constraints $x_j + x_{j'} \ge 2 + \delta$ gives $\sum x_j \ge 3 + \frac{3\delta}{2} > 3 + \delta = \mathrm{OPT}$, so the transferable core is empty; the allocation (batch $\{1, 2\}$, order 3 individually) is $\theta = 0$-unblocked since every other batch caps a covered order at a value already attained. □

Complexity. NP-hardness is a reduction from 3-dimensional matching: orders are ground elements, each element has an individual bid of value 1, and each triple is represented by a solver whose singleton values on the three covered elements are all 1 and whose triple bid produces $1 + \delta/3$ on each of the three orders (total value $3 + \delta \le 3g$ for $\delta \le 3(g-1)$). Thus the triple satisfies the complementarity bound and passes the fairness filter. If the maximum number of disjoint triples is m, the optimum welfare is $n + \delta m$, where n is the number of ground elements; hence welfare $n + \delta n/3$ is attainable iff a perfect matching exists. For fixed-parameter tractability, treat each directed pair as an atomic item under UDP: all orders on the same directed pair settle together, so a feasible solution either assigns the entire directed-pair block to one bid or leaves it unserved. After fair filtering, every surviving bid is a weighted subset of the P atoms, and winner determination is weighted set packing on a universe of size P. The dynamic program $DP[M] = \max\{DP[M \setminus T_b] + w_b : b \text{ uses } T_b \subseteq M\}$ over masks $M \subseteq [P]$ scans the bid list at each state, giving $2^{O(P)} \cdot \mathrm{poly}(|\mathcal{B}|)$.

C Full Proofs for Section 5

Proof (Proof of Theorem 4). Let there be one order. Solver i has private deliverable value v_i and may report any binding value $r_i \le v_i$; over-reporting is excluded by settlement, while under-reporting is feasible. By non-wastefulness the order is allocated whenever some report is positive, and by value-respecting the winner has a maximal report. Feasibility implies the winner can deliver at most its true value. Best-bid fairness requires the delivery to be at least the endogenous benchmark $\max_i r_i$.

Let the highest and second-highest true values be $v^{(1)} > v^{(2)}$. If the top solver reports truthfully, it sets the benchmark to $v^{(1)}$, must deliver $v^{(1)}$, and earns zero surplus. If instead it reports $r = v^{(2)} + \varepsilon$ with $0 < \varepsilon < v^{(1)} - v^{(2)}$, it still has the maximal report, the benchmark falls to $v^{(2)} + \varepsilon$, and it can earn $v^{(1)} - v^{(2)} - \varepsilon > 0$. Truthful reporting is therefore not a dominant strategy for any feasible non-wasteful value-respecting mechanism enforcing the best-bid benchmark. The same argument shows that any DSIC mechanism in this class can enforce at most the runner-up benchmark. ☐

Proof (Proof of Theorem 5). Lower bound. Restrict to the screening family R_g: a single strategic solver holds one batch of private value $V \in [1, g]$ (floor mass normalized to 1, so $\mathrm{FAIR}(V) = V$), and the mechanism posts an outcome as a function of the reported V. Write $p(V)$ for the probability the batch is executed and $x(V)$ for the expected batch delivery (zero when not executed). The fallback floor delivers welfare 1, so $W(V) = x(V) + 1 - p(V)$, and the solver's interim utility is $u(V) = p(V)V - x(V)$. Because bids are binding, a type V can only *under*report (claim $V' \le V$), so incentive compatibility implies

$$u(V) \ge u(V') + (V - V')p(V') \qquad (V' \le V).$$

Thus u is nondecreasing and $u'(V) \ge p(V)$ a.e. Hence $u(V) \ge \int_1^V p(t)\,dt$ and

$$x(V) = p(V)V - u(V) \le \psi(V) := p(V)V - \int_1^V p(t)\,dt.$$

Integrating $\psi(V)/V^2$ over $[1, g]$ and applying Fubini gives

$$\int_1^g \frac{\psi(V)}{V^2}\,dV = \int_1^g \frac{p(V)}{V}\,dV - \int_1^g p(t)\Big(\frac{1}{t} - \frac{1}{g}\Big)\,dt = \frac{1}{g}\int_1^g p(t)\,dt \le \frac{g-1}{g}.$$

If the mechanism has competitive ratio r, i.e. $W(V) \ge V/r$ pointwise, then $x(V) \ge V/r - (1 - p(V)) \ge V/r - 1$, and

$$\int_1^g \frac{V/r - 1}{V^2}\,dV = \frac{\ln g}{r} - \Big(1 - \frac{1}{g}\Big).$$

Combining the two inequalities, $(\ln g)/r \le 2(1-1/g)$, so every randomized DSIC mechanism has

$$r = \sup_{V\in[1,g]} \frac{V}{W(V)} \ge \frac{g \ln g}{2(g-1)} \ge \frac{\ln g}{2}.$$

Posted-price upper bound. Offer the batch a take-it-or-leave-it delivered welfare on the geometric ladder $\{e^k : 0 \leq k \leq K\}$, $K = \lceil \ln g \rceil$, choosing k uniformly at random; otherwise settle at floors. For any realized $V \in [e^k, e^{k+1})$, the draw k (probability $1/(K+1)$) executes the batch and delivers at least $e^k > V/e$, while all draws deliver at least the floor mass. Hence $\mathbb{E}[\text{welfare}] \geq \max\big(V/(e(K+1)), 1\big) \geq \text{FAIR}/(e(\lceil \ln g \rceil + 1))$.

Deterministic optimum. Let the mechanism be deterministic. First, its execution rule is a threshold: if $p(r) = 1$ and $p(V) = 0$ for some $r < V$, type V has truthful utility 0, while under-reporting r yields utility $V - x(r) \geq V - r > 0$ (feasibility at report r gives $x(r) \leq r$), contradicting downward IC. Thus there is a threshold τ (up to the boundary point) such that types below τ are not executed and types above τ are. Second, the accepted delivery is bounded by τ: for $\tau \leq r < V$, downward IC gives $V - x(V) \geq V - x(r)$, hence $x(V) \leq x(r)$; letting $r \downarrow \tau$ and using feasibility $x(r) \leq r$ gives $x(V) \leq \tau$ for all executed types, in particular $x(g) \leq \tau$. Types just below τ have welfare 1, while type g has welfare at most τ, so every deterministic DSIC rule has ratio at least $\max\{\tau, g/\tau\} \geq \sqrt{g}$. The posted delivery $\sqrt{g}$ attains this bound. Since R_g embeds as a degenerate single-minded subfamily of the full model, all three bounds transfer. □

Proof (Proof of Theorem 6). (i) Eligibility-posted packing. In the public-floor regime, draw $k \sim \text{Unif}\{0, \ldots, K\}$, $K = \lceil \ln g \rceil$, post the weight $C_S e^k$ for each option S, and let a solver accept its option iff its binding value clears this weight; among accepted options take a maximum-weight disjoint packing using the posted weights. Accepted options deliver their posted weights componentwise by the filter-passing scaling condition from the mechanism model; all orders not covered by the selected posted options settle at their public floors. The posted weights and packing rule are report-independent, so each solver faces a take-it-or-leave-it offer and the mechanism is universally truthful. Let X be the total value of the batch options used by the offline fair optimum. The geometric draw captures expected posted delivery at least $X/(e(K+1))$. Independently, every realization settles all orders at least at their public floors, so $W \geq C$. Since $\text{FAIR} \leq X + C$, the max-inequality with $a = e(K+1)$ gives $\mathbb{E}[W] \geq \text{FAIR}/(e(\lceil \ln g \rceil + 1) + 1)$.

(ii) Coupled menu. Draw k uniformly from $\{0, \ldots, K\}$ and set $t = e^k$. Post every option at price $C_S\, t$. Let the solver demand a profit-maximizing disjoint collection $\mathcal{C}^*(t)$. This is universally truthful because the posted menu is report-independent. It is feasible because any demanded S has $V_S \geq C_S t \geq C_S$, and the filter-passing scaling condition from the mechanism model implements total delivery $C_S t$ while clearing floors componentwise. Let

$$p(t) = \max_{\mathcal{C} \text{ disjoint}} \sum_{S \in \mathcal{C}} (V_S - C_S t).$$

As a maximum of finitely many affine decreasing functions, p is convex, piecewise linear, nonincreasing, with $p(t) = 0$ for $t \geq g$ (every option has $V_S \leq g C_S$). At any differentiability point $-p'(t) = \sum_{S \in \mathcal{C}^*(t)} C_S$, so the delivered mass is $D(t) = t \sum_{S \in \mathcal{C}^*(t)} C_S = t\,(-p'(t))$; at a kink the right-derivative selects the continuing (smallest-floor) active collection, and $D(t) \geq t\,(-p'_+(t))$. On $[t_k, t_{k+1}] = [e^k, e^{k+1}]$, convexity gives $-p'$ nonincreasing, so

$$\int_{t_k}^{t_{k+1}} (-p'(t))\,dt \leq (-p'_+(t_k))(t_{k+1} - t_k) = (-p'_+(t_k))(e-1)e^k \leq (e-1)D(t_k).$$

Summing over $k = 0, \ldots, K-1$ and using $p(e^K) = 0$,

$$p(1) = p(1) - p(e^K) = \int_1^{e^K} (-p'(t))\,dt \leq (e-1)\sum_{k=0}^{K} D(e^k),$$

so the uniform draw gives $\mathbb{E}[D] \geq p(1)/((e-1)(K+1))$. Since the fair optimum's batches form a feasible disjoint collection at $t = 1$, $p(1) \geq \sum_{S\in B^*}(V_S - C_S) = \mathrm{FAIR} - C$, and combining with the pointwise $W \geq C$ via the same max-inequality yields $\mathbb{E}[W] \geq \mathrm{FAIR}/((e-1)(\lceil \ln g \rceil + 1) + 1)$.

Independent geometric pricing fails. The separation uses the following construction. There are options $S_t = \{0, t\}$ for $t = 1, \ldots, m$, all sharing the common order 0, with $c_0 = c_t = \frac{1}{2}$ and $C_{S_t} = 1$. The solver has value $V_t = g$ for every option. Independent geometric prices make the solver choose only the cheapest acceptable option: $D = P_{\min}$ if $P_{\min} \leq g$, and $D = 0$ otherwise. Hence

$$D \leq 1 + g\,\mathbf{1}\{P_{\min} \geq e\}.$$

Moreover,

$$\Pr[P_{\min} \geq e] = (K/(K+1))^m \leq e^{-m/(K+1)}.$$

Choosing $m = \lceil (K+1)\ln g \rceil$ gives $\mathbb{E}[D] \leq 2$. Thus $\mathbb{E}[W] \leq 2 + C = O(\log^2 g)$, while the fair optimum obtains $\mathrm{FAIR} = g + (m-1)/2 = \Omega(g)$ by taking one high-value option and settling the remaining leaf orders at floors. Hence the independent geometric analogue has ratio $\Omega(g/\log^2 g)$. □

Proof (Proof of Theorem 7). Upper bound. In random-priority coupled menus, condition on a solver being served first (probability $1/N$): it faces the full coupled menu on all orders, and the profit-curve telescope of Theorem 6(ii) applies verbatim, giving expected delivery at least $p_i(1)/((e-1)(K+1))$ from that solver, where p_i is computed on the full order set. A solver served later faces a subset of orders and contributes nonnegatively. Since each solver's optimal batches form a disjoint collection on the full set, $\sum_i p_i(1) \geq \sum_{b\in B^*}(V_b - C_b) = \mathrm{FAIR} - C$. Averaging over the uniform priority and combining with $W \geq C$ gives $\mathbb{E}[W] \geq \mathrm{FAIR}/(N(e-1)(\lceil \ln g \rceil + 1) + 1)$. Truthfulness holds because a solver's menu depends only on which orders predecessors have consumed, never on its own report.

Lower bound (symmetrized swarm). Fix orders $o_1, \ldots, o_N$ with floors $\varepsilon = L/(N\rho)$ and auxiliary orders $u_1, \ldots, u_L$ with floor 1; let G be the set of all orders, so $C_G \leq 2L$. Each identity has the *same* public option family $\{\{o_i\}, G\}$; identities differ only in private values, so any report-independent service order is statistically independent of which identity is the "batcher." Choose $\rho \in (\max\{1, g/2\}, g]$

avoiding the (countably many) atoms of the N multiplier distributions. The batcher, placed by the adversary at an identity chosen after seeing the mechanism, has $V(G) = \rho C_G$ and a ratio-1 singleton; every other identity (a blocker) has a ratio-ρ singleton on its own order and a ratio-1 copy of G. At multiplier t a blocker is silent iff $t > \rho$; write $q_i = \Pr[t_i > \rho]$. The batcher delivers $\Theta(\rho C_G)$ only if it is reached before any blocker has consumed an order of G, i.e. only if all predecessors are silent. Blockers together deliver at most $N\varepsilon\rho = L$, and fallback floors contribute at most $C_G \le 2L$, so pointwise $W \le \mathrm{del}_{\mathrm{batcher}} + 3L$. For any realization of the order,

$$\sum_i (1 - q_i) \prod_{s \prec i} q_s \; = \; 1 - \prod_s q_s \; \le \; 1,$$

the i-th term being the probability that i is the first non-silent identity. Taking expectation over the (independent) order and the multipliers,

$$\sum_i (1 - q_i) \Pr[\text{predecessors of } i \text{ silent}] \le 1,$$

so some identity i^* has $(1 - q_{i^*}) \Pr[\text{predecessors silent}] \le 1/N$. Placing the batcher at i^* caps its expected batch delivery at $2\rho L/N$, while the fair optimum delivers at least ρL via the batch. Hence

$$\frac{\mathrm{FAIR}}{\mathbb{E}[W]} \ge \frac{\rho}{3 + 2\rho/N} = \Omega(\min(N, \rho)) = \Omega(\min(N, g)),$$

using $\rho > g/2$. The blockers' geometry is public and identical across placements, so the bound holds even for geometry-aware orders. □

D Full Proofs for Section 6

Proof (Proof of Proposition 3). Settle-at-deadline is fair and achieves C on every instance, while $\mathrm{FAIR} \le \mathrm{OPT} \le gC$, so the ratio is at most g; a staggered-deadline instance forcing individual-only settlement shows it is at least g. □

Proof (Proof of Theorem 8). (i) If an almost-surely fair mechanism settles an exposed batch (a covered order's deadline in the future) with positive probability on some prefix, fix that prefix and append a higher standalone bid on a later-deadline covered order: on the appended instance it has, with positive probability, settled below a realized floor—not a.s. fair. Hence exposed batches are never settled, and on the staggered hard family the mechanism is individual-only a.s., with welfare $\le C$ against $\mathrm{FAIR} = gC$. (ii) Stopped welfare $q = V + C - C_S$ satisfies $q \in [C, gC]$ ($V \ge C_S$ and $V \le gC_S$, $C_S \le C$). *Deterministic* $\Theta(\sqrt{g})$: accept the first $q \ge \sqrt{g}\,C$; if $q^* \ge \sqrt{g}\,C$ the ratio is $q^*/q \le \sqrt{g}$, else $W = C$ and ratio $< \sqrt{g}$; a two-step adversary matches. *Randomized* $O(\log g)$: with $k \sim \mathrm{Unif}\{0, \dots, K\}$, accept the first $q \ge e^k C$. With probability $1/(K + 1)$, $e^k C \le q^* < e^{k+1} C$, and then the mechanism accepts some offer with $q \geq e^k C > q^*/e$, so $\mathbb{E}[W] \geq q^*/(e(K+1))$. The matching $\Omega(\log g)$ is a Yao bound: a persistent order with fleeting geometric batch offers $x_i = e^i c$, $i \leq \lfloor \ln g \rfloor$, shown for the first M with $\Pr[M \geq i] = e^{-i}$, caps every deterministic threshold at $2c$ while $\mathbb{E}[\mathrm{FAIR}] = \Omega(c \log g)$. Greedy: a ratio-$g$ blocker arriving before a ratio-g batch is accepted (killing the batch) for any threshold $T \leq g$ and rejected with the batch for $T > g$, ratio $\Omega(g)$. □

Proof (Proof of Theorem 9). In the per-order-decomposable individual game, the maximal standing bid on j both wins and delivers itself, so $c_j(b) = p_j(b)$. Let $i^*(j)$ deviate to W_j with density $1/(v_j^{\max} - w)$ on $[0, (1-1/e)v_j^{\max}]$ (Syrgkanis–Tardos). Then the expected utility from j is $((1-\frac{1}{e})v_j^{\max} - \beta_j)^+ \geq (1-\frac{1}{e})v_j^{\max} - p_j(b)$ with $\beta_j \leq c_j(b) = p_j(b)$, the penalty landing under b. Summing and using $\sum_i u_i(b) = \mathrm{PW}(b) - \sum_j p_j(b)$ in the CCE inequality cancels the price terms, giving $\mathbb{E}[\mathrm{PW}] \geq (1-1/e)\mathrm{OPT}_{\mathrm{ind}}$; against the unrestricted optimum, $\mathrm{OPT} \leq g\,\mathrm{OPT}_{\mathrm{ind}}$, so the factor is at most $(e/(e-1))g$. The pure-NE lower bound is Theorem 1's instance with a filtered $(2gv, 0)$ batch. The full-game obstruction is a latent batch present in b_{-i} that executes under the deviation but not under b, breaking every charging scheme. □

E Benchmark Manipulation

On one directed pair, let an attacker of volume v depress a quantile benchmark by injecting mass a at convex cost $K(a) = \kappa_1 a + \frac{\kappa_2}{2}a^2$, with local sensitivity β. For a lower-tail quantile of m honest quotes with density f at b^*, first-order sensitivity is $\beta = 1/(mf(b^*))$.

Theorem 10 (The contamination game). *A marginal injection cost $\kappa_1 \geq v\beta$ deters contamination. Otherwise the optimal steady attack is $a^* = \min\{\bar{a}, (v\beta - \kappa_1)/\kappa_2\}$, inducing relative bias $\mu = \beta a^*/b^*$; bursts do not beat the constant attack, and rate limits tax restarts through a finite ramp without changing the steady state. Every floor-calibrated guarantee is then read with $g \mapsto g/(1-\mu)$.*

Proof (Proof of Theorem 10). The steady-state benchmark under a constant injection a is $b^* - \beta a$ (the rate limit does not bind once the bias is constant), so the per-round profit is $\pi(a) = v\beta a - \kappa_1 a - \frac{\kappa_2}{2}a^2$. With $\kappa_2 > 0$, π is strictly concave with $\pi'(0) = v\beta - \kappa_1$; hence if $v\beta \leq \kappa_1$ the unique maximizer is $a^* = 0$ (no attack), and otherwise $a^* = \min\{\bar{a}, (v\beta - \kappa_1)/\kappa_2\}$, with relative bias $\mu = \beta a^*/b^*$.

Bursts do not help. The published benchmark satisfies $b_t \geq b^* - \beta a_t$ pointwise, so for every horizon T and every (possibly time-varying) policy (a_t),

$$\sum_{t=1}^{T} \big[v(b^* - b_t) - K(a_t)\big] \leq \sum_{t=1}^{T} \big[v\beta a_t - K(a_t)\big] = \sum_{t=1}^{T} \pi(a_t) \leq T\,\pi(a^*),$$

and the constant policy a^* attains $T\pi(a^*) - O(1)$ after the ramp; hence no episodic schedule beats the constant attack.

Ramp tax (exact). Starting from $b_0 = b^*$, the rate limit caps the attainable bias at round t by $\Delta_t := \min\{\Delta^*, b^*(1-(1-r)^t)\}$ with $\Delta^* = \beta a^*$; reaching Δ^* takes $T_\mu = \lceil \ln \frac{1}{1-\mu} / \ln \frac{1}{1-r} \rceil$ rounds. An attacker who hugs the cap injects $a_t = \Delta_t/\beta$, so the exact net rent forgone per restart is

$$L_{\text{net}} = \sum_{t=1}^{T_\mu} \Big[\pi(a^*) - \pi(\Delta_t/\beta)\Big] \;\le\; \pi(a^*)\, T_\mu.$$

In the small-(r,μ) regime, $1-(1-r)^t \approx rt$ and the summand is $\frac{\kappa_2}{2}(a^*)^2(1-rt/\mu)^2$, giving $L_{\text{net}} = \frac{\pi(a^*)\mu}{3r}(1+o(1))$; we state this only as a first-order approximation, the exact figure being the finite sum above.

Feedback. A steady bias μ scales every floor on the pair by $1-\mu$, so any guarantee whose benchmark input is the public-floor spread $G = \max_{i,S:\, C_S>0} V_i(S)/C_S$ becomes $G/(1-\mu)$; the coverage law transforms jointly, $(g,\alpha) \mapsto (g/(1-\mu), \min\{1, \alpha/(1-\mu)\})$. Structural production complementarity, directed-pair complexity, and produced-welfare PoA are unaffected. $\square$

F Deployment Diagnostics

Table 2. Deployment diagnostics used only as motivation. The visible spread is auction-level and observable; it is not the structural complementarity factor g.

Diagnostic	Value or interpretation
Competitions in trace	802,816 post-adoption solver competitions
Covered notional / gap	\$54.69B / \$5.13M (about 0.94 bps)
Visible spread tail	$\widehat{G}^{\text{obs}} > 2$: 25.3%; $\widehat{G}^{\text{obs}} > 4$: 20.2%
Small-notional tail	$\widehat{G}^{\text{obs}} > 2$: 63.1%
Directed-pair support P	directly observable from requested/paid pairs
Naive local reserve	97.1% vulnerable cell-windows in diagnostic

Sample and filters. The trace contains 802,816 post-adoption solver competitions over an eight-month window. The parser retains protocol-valid, non-filtered, positive-score solver submissions. Full-sample quantities such as visible spread and effective solver count use all parsed auctions. USD-denominated quantities use smaller denominators because score conversion, notional coverage, and sanity filters are available only for subsets of the trace. These denominator changes are coverage restrictions, not inconsistencies.

Score variables. For each competition, Score_1 is the winning solution score. The reference score $\text{Score}_2^{\text{other}}$ is the best positive score submitted by a solver different from the winner. The parser groups by solver identity, removes all candidate solutions submitted by the winning solver, and then takes the maximum remaining positive valid score; if none exists, the reference score is zero. This best-non-winning construction is solver-level: a winning solver's second candidate solution is not a credible outside option.

Visible spread and accounting gap. When the denominator is positive, $\widehat{G}^{\text{obs}} = \text{Score}_1/\text{Score}_2^{\text{other}}$, with infinite spread when the reference score is zero. The reported tail shares are computed on the full 802,816-auction sample. The dollar gap converts $\text{Score}_1 - \text{Score}_2^{\text{other}}$ using ETH/USD prices, winsorizes the dollar gap at the first and ninety-ninth percentiles, and sums over the sanity-filtered denominator above. The $54.69B covered notional is the sell-token notional over the same denominator.

Reserve-vulnerability diagnostic. The benchmark diagnostic uses cell-windows from the reserve-construction pipeline. With baseline quantile $\tau = 0.25$, a cell-window is marked vulnerable if the maximum eligible-sample share of any single solver is at least τ, meaning that a single solver could potentially influence the local reserve quantile if its samples were contaminated. In the diagnostic, 97.1% of auction-weighted cell-windows are vulnerable. The design implication is to inherit from parent cells or use audit-only/soft reserves when local support is sparse.